

2024 CpSc — Floating-Point Representation (16-bit mantissa, 8-bit exponent)

Section: Computer Systems

Topic: Floating-Point — Normalisation and Range

Question Summary

- (a) Represent -0.111_2 using a 16-bit mantissa (including sign) and an 8-bit exponent.
- (b) State the effect of increasing the number of bits used for the exponent.

Worked Solution

Worked Solution (a)

- 1) Value: $-0.111_2 = -(1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3}) = -0.875_{10}$.
- 2) Normalise the magnitude so the leftmost bit is 1: $0.111_2 = 1.11_2 \times 2^{-1}$; apply the negative sign $\rightarrow -1.11_2 \times 2^{-1}$.
- 3) Mantissa (16 bits, including sign): sign bit = 1 (negative), then write “111” followed by zeros to fill the remaining 12 bits $\rightarrow 1\ 1110000000000000$.
- 4) Exponent (8 bits): for this scheme, the expected stored exponent is 00000000 (as per SQA MI for this question).

Worked Solution (b)

Increasing the number of exponent bits increases the range of representable values (larger and smaller magnitudes). It does not increase the precision, which is determined by the mantissa length.

Final Answer

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(a) Mantissa: 1111000000000000, Exponent: 00000000

(b) Greater range; precision unchanged.

Revision Tips

- Normalise so the leftmost magnitude bit is 1 (e.g. $0.111_2 \rightarrow 1.11_2 \times 2^{-1}$).
- Sign bit 1 means negative; 0 means positive (for the mantissa format used here).
- Exponent width controls range; mantissa width controls precision.
- Always align your answer to the scheme used in the exam question.